

BIT 4107-Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

Student name: KELLY MAROA

Admission Number:BIT/2024/49929

School/Faculty: COMPUTING AND INFORMATICS

Programme:Blood_Donation_App

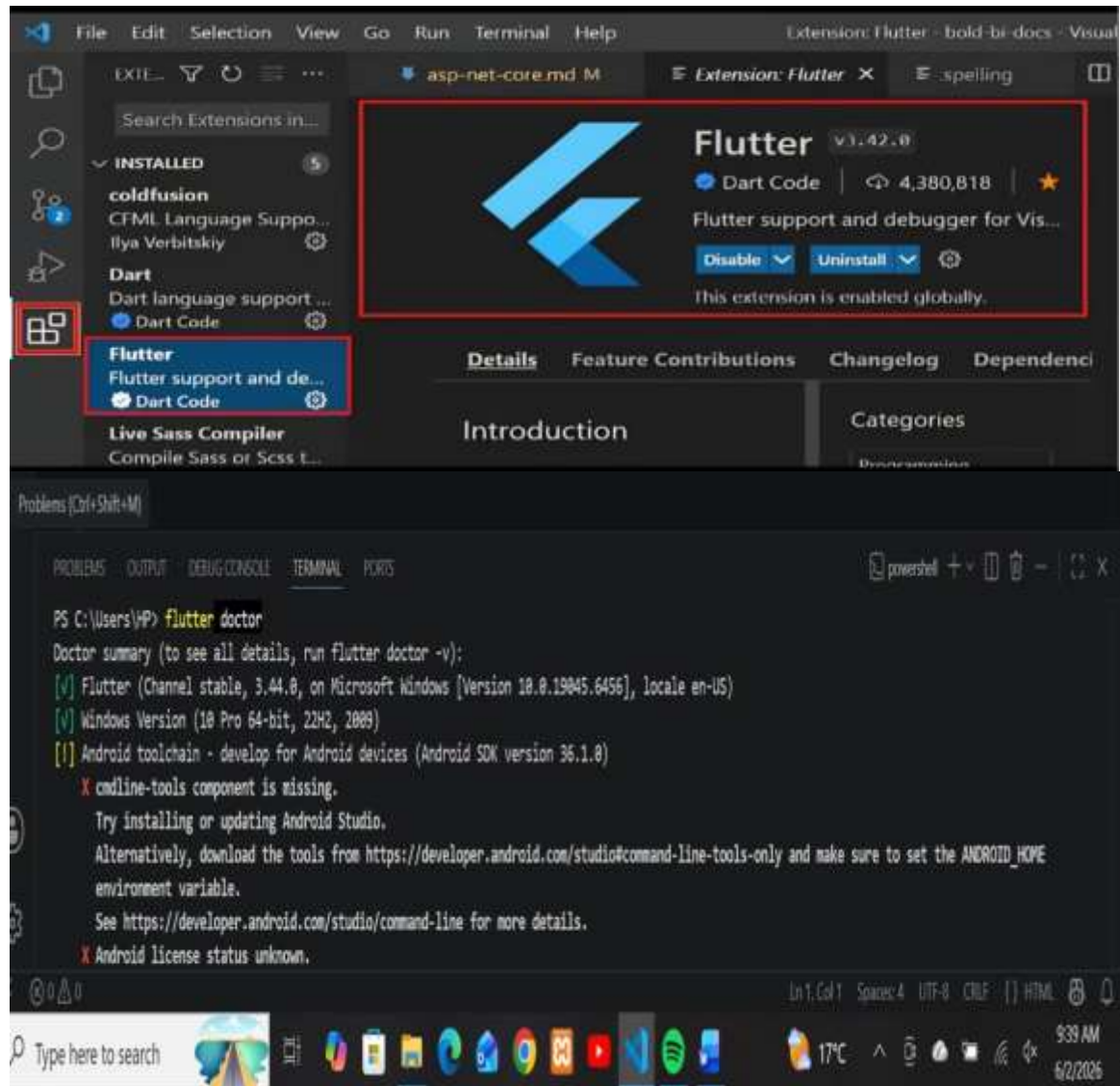
Academic Year: THREE

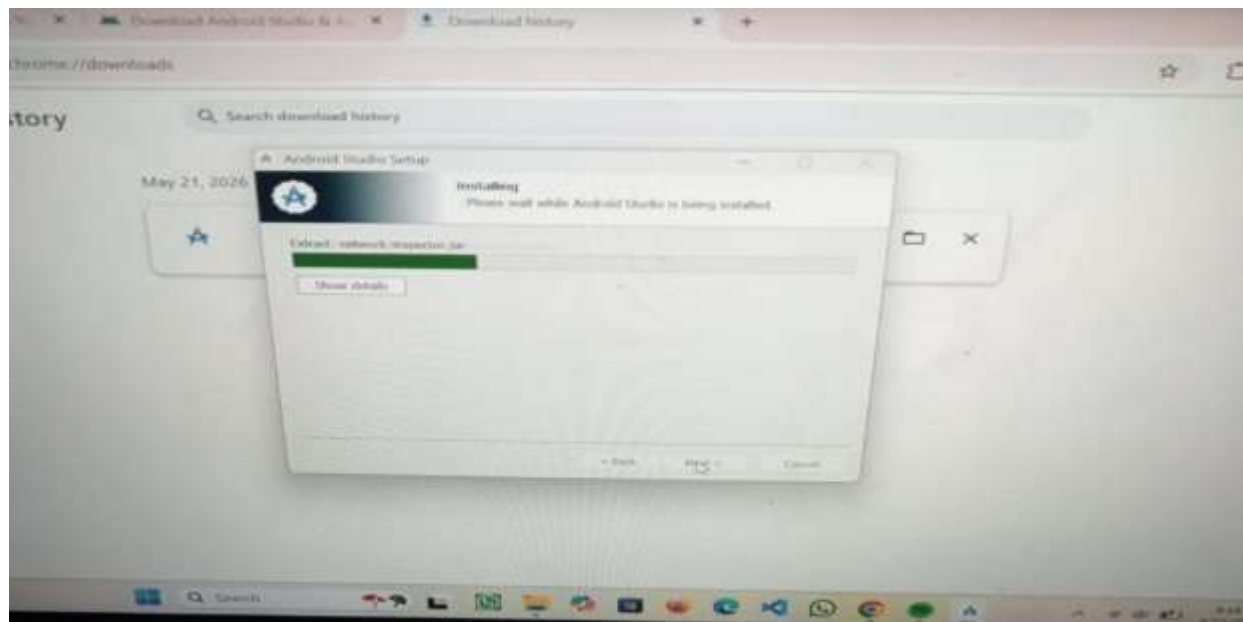
Semester:TWO

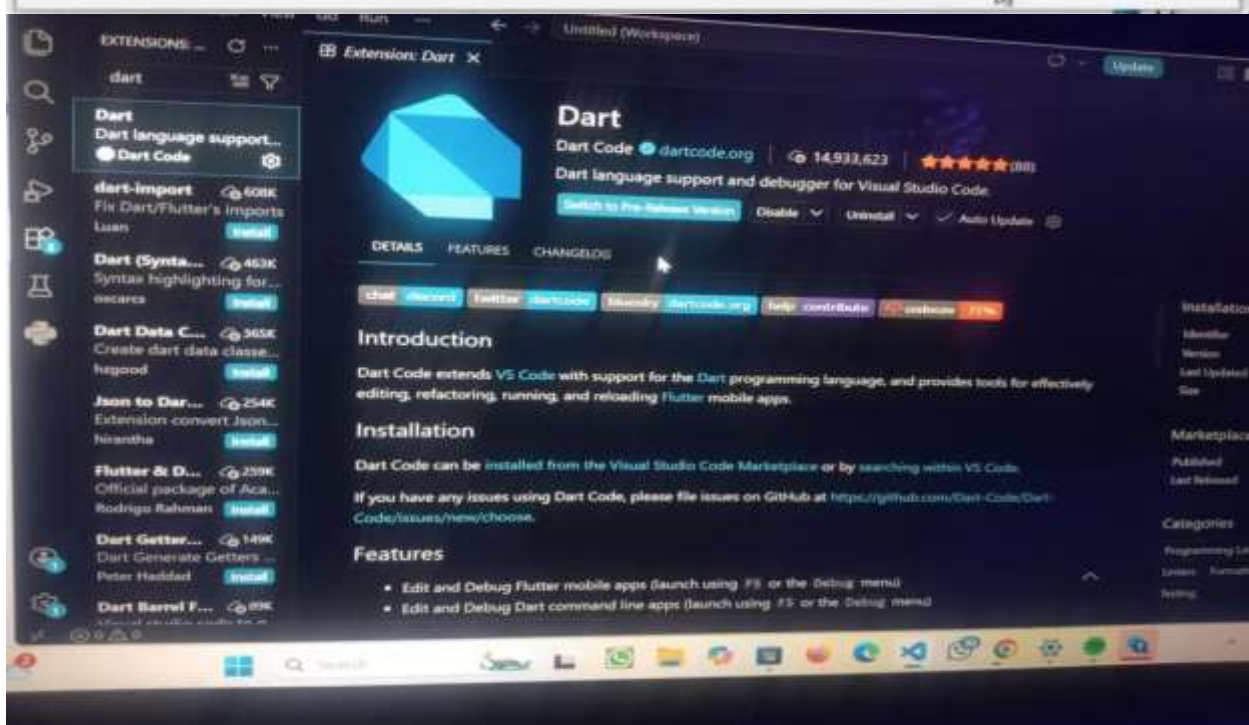
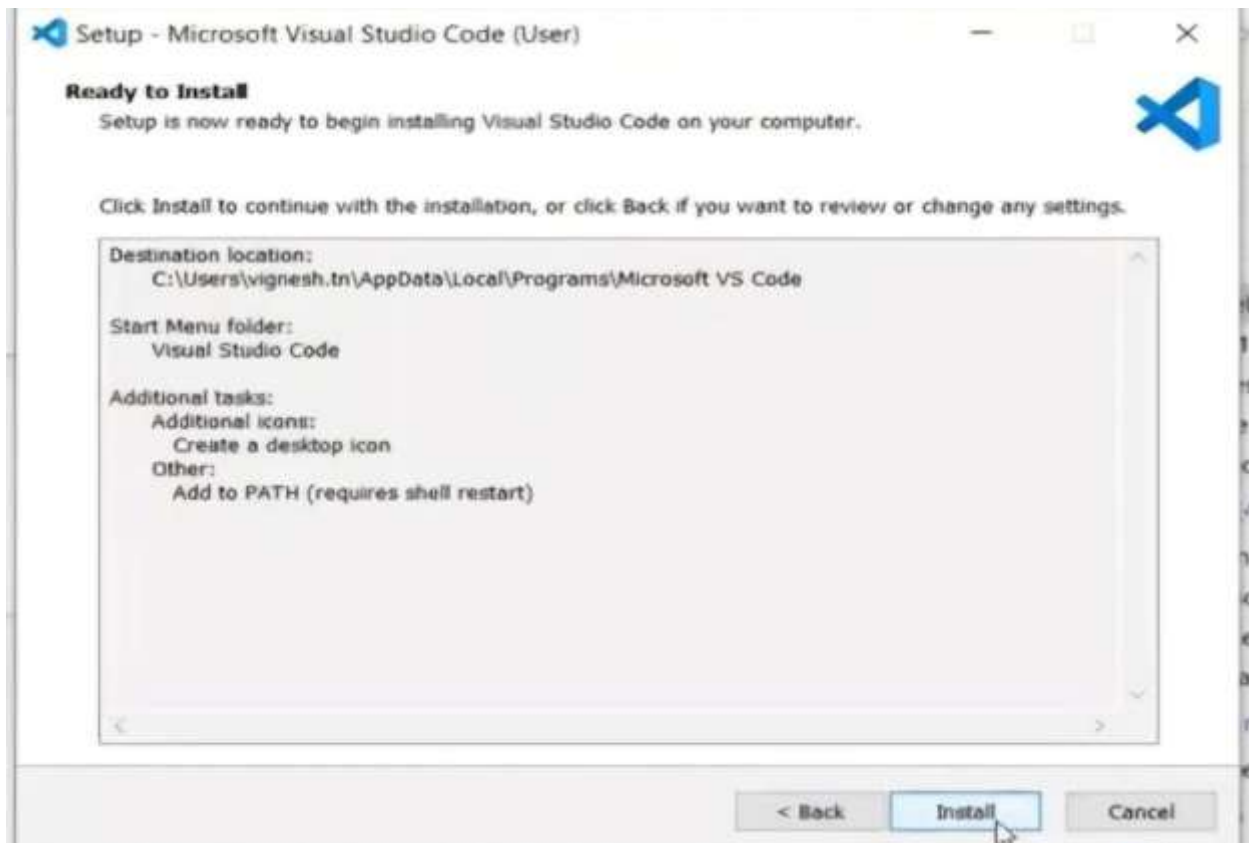
Lecturer: MR.MICHAEL NYORO

Phone Number:0725104634

WEEK 1: INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT







TOOLSANDSOFTWAREUSED 1.Android
Studio

2.VS Code

3.Flutter SDK

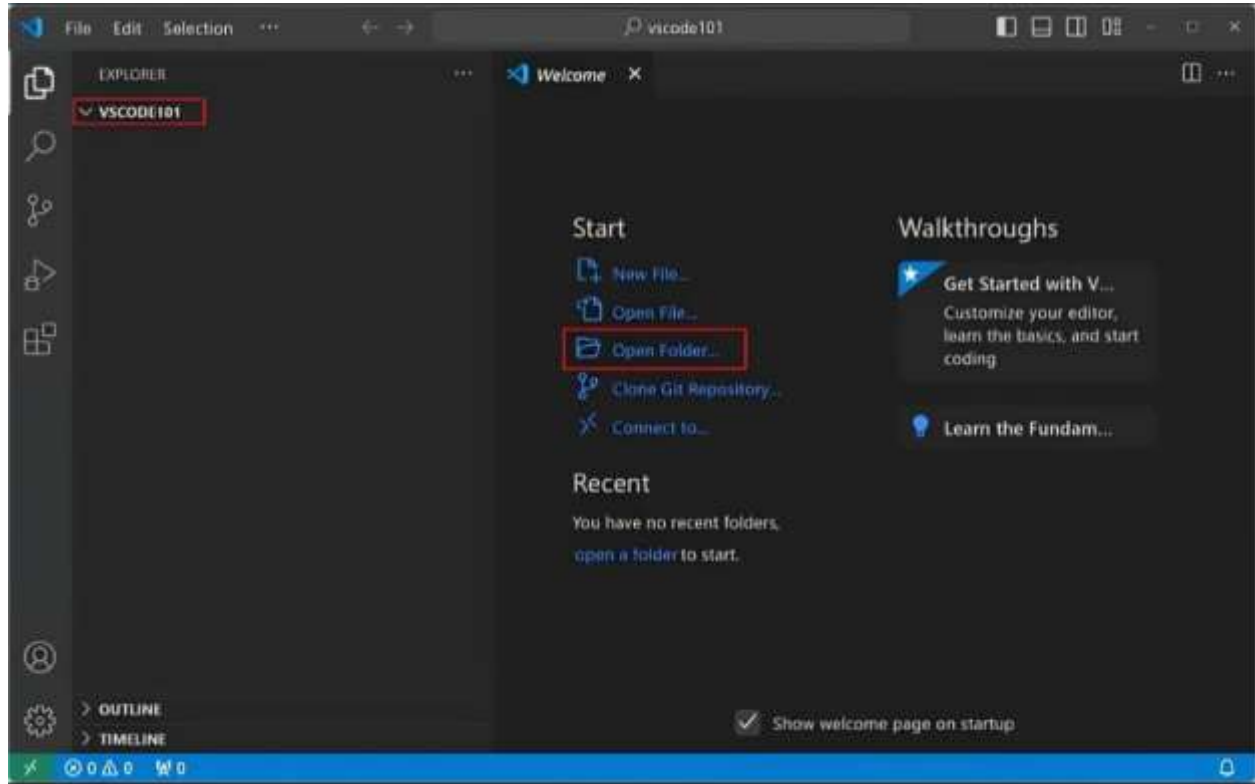
4.Dart

5.Android Phone

PERSONAL REFLECTION

Installing the tools was challenging at first due to errors and configuration issues, but it improved my problemsolving skills. I gained practical experience in setting up development environments and became more confident in using modern software tools for app development.

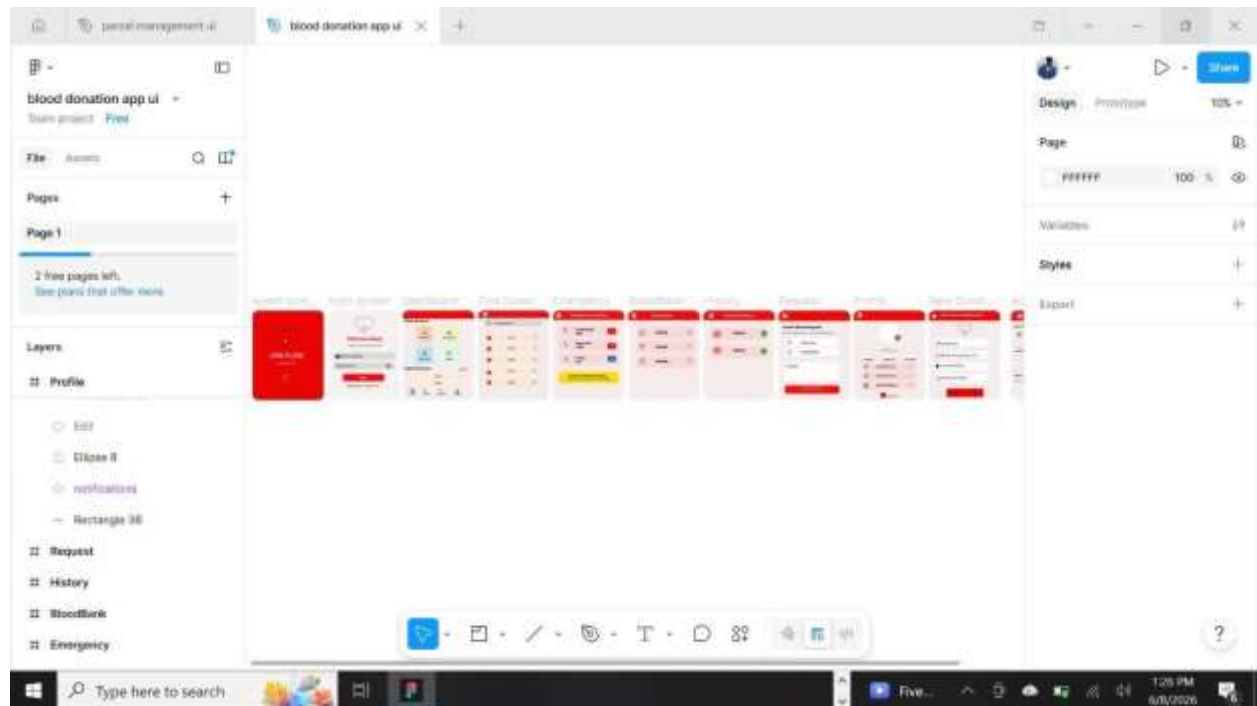
WEEK 2: MOBILE DEVELOPMENT ENVIRONMENT



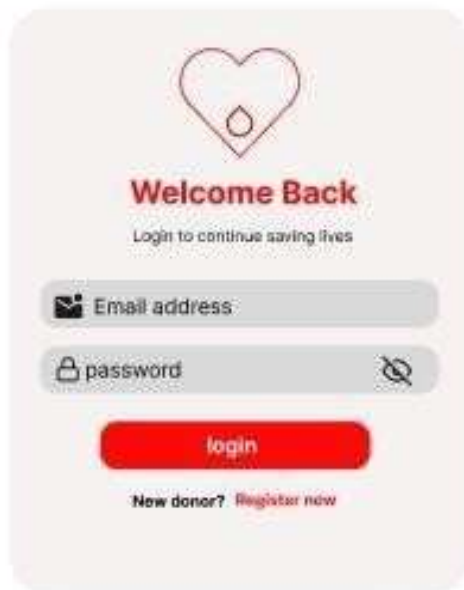
PERSONAL REFLECTION

I set up the mobile development environment by installing and configuring tools like Android Studio, Flutter SDK, and device emulators. This process helped me understand how different components work together, though I faced minor setup challenges that enhanced my troubleshooting and technical skills.

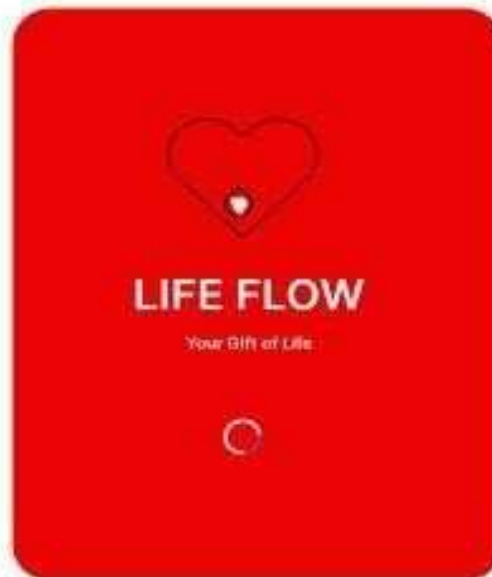
WEEK 3: USER INTERFACE DESIGN



login screen



spash screen

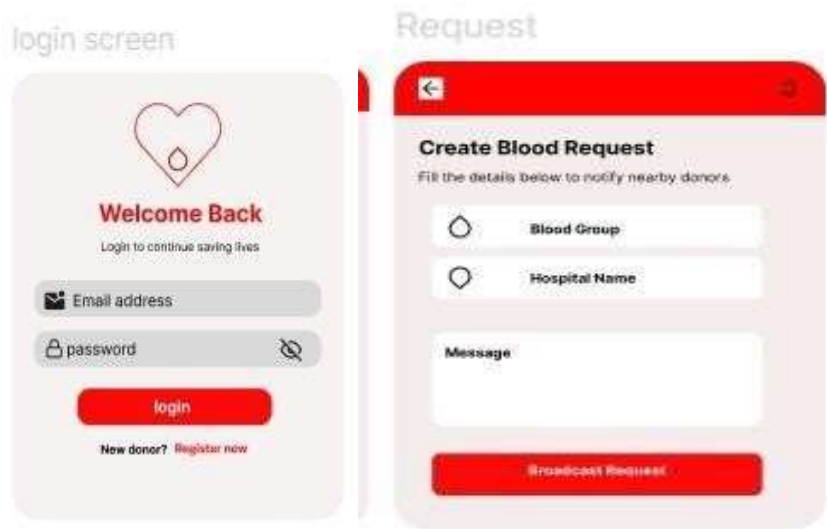


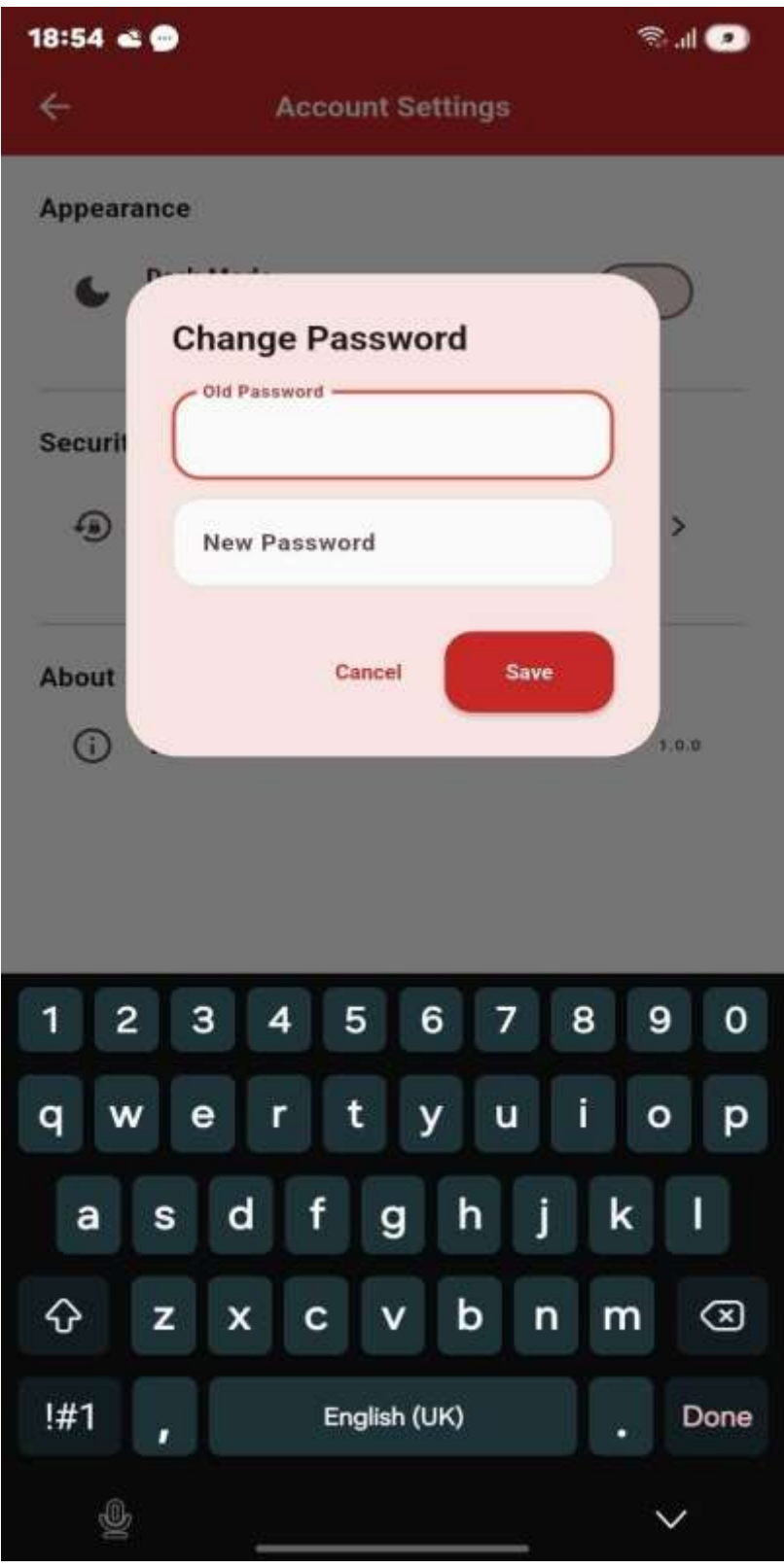


Personal reflection

I focused on user interface design for the blood donation app using Figma. I learned how to create and organize screens, design user-friendly layouts, and improve overall system usability. This improved my understanding of UI/UX principles and enhanced my design skills.

Week 4: Event handling and user interaction





Personal reflection

I learned about event handling and user interaction in the blood donation app. I implemented how users interact with the system through buttons, forms, and input fields. This improved my understanding of how applications respond to user actions and enhanced my programming skills.

Week 5: Data management

19:08



New Donor Registration



Full Name



Kelly maroa

Blood Group (e.g. O+)



O+

Location/City



Machakos

Phone Number



0785324646

Register as Donor



Search Blood Group

**Anna Itotiah W.**

Nairobi

**Sharon Kendi.**

Nakuru

**Faith Mueni.**

Mombasa

**Roy Gichinga.**

Kisumu

**Michelle Njogu.**

Eldoret

**Kelley Maroa**

Migori

**Nicole Chepkemboi**

Nandi



Home



Search



Request



Profile



Personal reflection

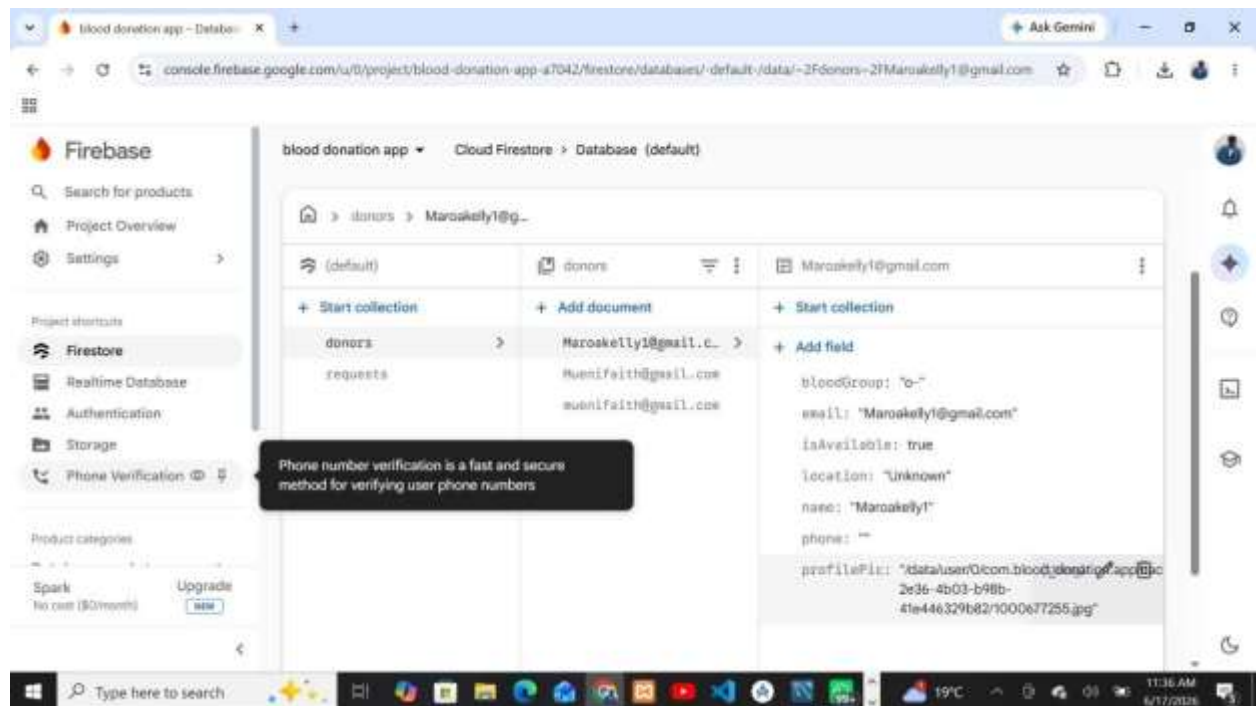
I focused on data management in the Blood Donation App. I learned how to store, retrieve, and update donor and request information using local storage and databases. This improved my understanding of how data is handled efficiently and enhanced my skills in managing application data.

Week 6: Networking and APIs

```
1  {
2    "project_info": {
3      "project_number": "596793580149",
4      "project_id": "blood-donation-app-a7042",
5      "storage_bucket": "blood-donation-app-a7042.firebaseioapp"
6    },
7    "client": [
8      {
9        "client_info": {
10          "mobilesdk_app_id": "1:596793580149:android:d5f7eac4ef37ed8144363c",
11          "android_client_info": {
12            "package_name": "com.blood_donation.app"
13          }
14        },
15        "oauth_client": [],
16        "api_key": [
17          {
18            "current_key": "AIzaSyDVDy62fZd7rrAi_DwKwh7gyK9Nb91T9Wo"
19          }
20        ],
21        "services": {
22          "appinvite_service": {
23            "other_platform_oauth_client": []
24          }
25        }
26      }
27    ],
28    "configuration_version": "1"
29  }
```

A blood donation app requires networking to connect users with donors and hospitals, enable real-time updates, send requests, share location data, and store information on cloud servers for accessibility and communication.

Week 7: Data storage and Database integration in Mobile application



blood donation app - Database

console.firebase.google.com/u/0/project/blood-donation-app-a7042/firestore/databases/-default-/data/~2Frequests~2F56MW0kouZrbP2Ww...

blood donation app Cloud Firestore Database (default)

requests 56MW0kouZrbP2Wwgn8

+ Start collection + Add document + Start collection

docs requests pQicE7j+HQURL4dScwkd + Add field

blood: "D+"
date: "Just now"
location: "Agha Khan Hospital"
message: "Emergency"
serverTimestamp: 12 June 2026 at 13:03:15 UTC+3
timestamp: "2026-06-12T11:42:02.628035"
type: "Emergency Request"

Type here to search

19°C

11:37 AM 6/12/2026

blood donation app - Authen

console.firebase.google.com/u/0/project/blood-donation-app-a7042/authentication/users

blood donation app Authentication

Users Sign-in method Templates Usage Settings Extensions

The following authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Search by email address, phone number or user UID Add user

Identifier	Providers	Created	Signed in	User UID
muenlath@gmail.com		12 Jun 2026	12 Jun 2026	HxvzBzfy0rK5f0r3jHxK...
marcellely@gmail.com		11 Jun 2026	15 Jun 2026	cDM6nUQvR5dTSm2wB44X...

Rows per page: 50 1 - 2 of 2

Type here to search

19°C

11:38 AM 6/12/2026

- AB-

Agha Khan
Request - 2026-06-08


- B+

Kenyatta hospital
Request - 2026-06-05


- O+

Thika level 5
Request - 2026-06-05





Search Blood Group

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Nairobi

**Sharon Kendi.**

Nakuru

**Faith Mueni.**

Mombasa

**Roy Gichinga.**

Kisumu

**Michelle Njogu.**

Eldoret

**Kelley Maroa**

Migori

**Nicole Chepkemboi**

Nandi



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Search



Request



Profile

Personal reflection

Learning about data storage and database integration helped me understand how mobile apps manage and persist user data efficiently. I now appreciate the importance of choosing the right database and ensuring secure, reliable data handling in application development.

WEEK 8: Handling user inputs and Events

20:39



Welcome Back

Login to continue saving lives



Email Address

example@mail.com



Password



Maroakelly2
@gmail.com

urgent

o



20:40



Account Settings

Appearance



Dark Mode



Change Password

Old Password

New Password

Cancel

Save

Security



About



Version

1.0.0



20:39



Welcome Back

Login to continue saving lives



Email Address

example@mail.com

Enter a valid email



Password



Password must be 6+ characters

Login

New donor? [Register Now](#)

20:40



Hello, Maroakelly1



 Nairobi, KE (Lat: -1.19)

Quick Services



Find Donor



Emergency



Blood Bank



History

Smart Match Suggestion



Best Match Found!

Maro Kelly (O+) is 500m away and available.



Urgent Requests

[View All](#)

O+

Oasis
Just now
urgent!!



Home



Search



Request



Profile

20:40



Hello, Maroakelly1



Search Donor, Group or Location

mue



Faith Mueni

Machakos • AB-



muenifaith

Unknown • Unknown



mue

mueni

muen



20:45



← lib/main.dart



```
476         const SizedBox(height: 40),
477         Text(
478           "Welcome Back",
479           style: TextStyle(
480             fontSize: 32,
481             fontWeight: FontWeight.bold,
482             color: Colors.red.shade900,
483           ),
484         ),
485         const SizedBox(height: 10),
486         Text(
487           "Login to continue saving lives",
488           style: TextStyle(color: Colors
489             .grey.shade600, fontSize: 16),
490         ),
491         const SizedBox(height: 50),
492         TextFormField(
493           controller: emailController,
494           decoration: const InputDecoration(
495             labelText: "Email Address",
496             hintText: "example@mail.com",
497             prefixIcon: Icon(Icons
498               .email_outlined),
499             ),
500           keyboardType: TextInputType
501             .emailAddress,
502           validator: (v) => v!.contains('@') ?
503             null : "Enter a valid email",
504         ),
505         const SizedBox(height: 20),
506         TextFormField(
507           controller: passwordController,
508           obscureText: !isPasswordVisible,
509           decoration: InputDecoration(
510             labelText: "Password",
511             prefixIcon: const Icon(Icons
512               .lock_outline),
513             suffixIcon: IconButton(
514               icon: Icon(isPasswordVisible ?
```



Home



Inbox



Explore



Copilot

Personal reflection

This topic helped me understand how mobile apps handle user input through events and gestures and how to do validation. I learned to structure functionality using classes, making applications more organized, interactive, and easier to maintain while improving user experience.